

Theory of Stochastic Processes SS 2026, Exercise Sheet 1

15.04.2026

Homework due on Monday 20.04.2026, 8:30 AM

During each tutorial "Short questions" will be primarily discussed, as well as *some* of the homework questions due on that week (official solutions will be published eventually). Homework questions must be solved **individually** at home and uploaded online *before* the lecture on Monday morning of the following week. To qualify for the bonus system (+0.3 increase on the grade of **exclusively** the main exam, **not** the retake) at least 70% of the homework points have to be earned during the semester. Homework points for each task are awarded if at least a *genuine solution attempt* is shown by the student

For more info see: <https://academics.nat.tum.de/en/studies/org/mh/details/mod/PH1006>

Short questions

S 1 Probability Distribution Functions

- (a) How are the terms 'probability distribution' and 'probability density' related?
- (b) Does a probability distribution $P(x) : \Omega \rightarrow R^{0+}$ in general have to be smaller or equal than one, $P(x) \leq 1 \quad \forall x$? What if the state set is continuous? What if it is discrete?
- (c) Give an example of 2 random variables which are *uncorrelated* but **NOT** *independent*
- (d) Give some examples of probability distribution functions which appear commonly in physics

S 2 Means and Variances

- (a) Do the first and second moment of a probability distribution always exist?
- (b) Calculate mean and variance of the Exponential Distribution: $P(x) = \lambda \exp[-\lambda x]$ for $x \geq 0$
- (c) Calculate mean and variance of the Poisson Distribution: $P(n) = \frac{\lambda^n}{n!} e^{-\lambda}$ for $n \in N_0$

S 3 Definition of a Stochastic Process

- (a) Define a stochastic process.
- (b) Is a random variable a stochastic process?
- (c) Which of the following expressions for integrals over the one point probability distribution of a stochastic process is 1?

$$\int_{-\infty}^{\infty} P(x, t) dt \quad (1)$$

or

$$\int_{-\infty}^{\infty} P(x, t) dx \quad (2)$$

- (d) Now compare this to the joint probability distributions of two random variables X and Y , $P(x, y)$: What do the integrals

$$\int_{-\infty}^{\infty} P(x, y) dx \quad (3)$$

$$\int_{-\infty}^{\infty} P(x, y) dy \quad (4)$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} P(x, y) dx dy \quad (5)$$

give you?

- (e) How can you rewrite the joint 2-point probability distribution of a stochastic process $P(x_1, t_1; x_2, t_2)$ if the stochastic variables at t_1 and t_2 are independent of each other?

S 4 Characteristic Function and Cumulants

Let X_1 and X_2 be independent, real valued random variables with the characteristic functions $G_{X_1}(k)$ and $G_{X_2}(k)$. Let $Y = X_1 + X_2$.

- (a) Show that $G_Y(k) = G_{X_1}(k)G_{X_2}(k)$
- (b) Calculate the cumulants of Y as function of the cumulants of X_1 and X_2
- (c) Show that the first two terms of the cumulant's Taylor expansion encode the expectation value and the variance. Start from the definition $K_X(t) = \log M_X(t)$ and use $\log(1+x) = x - \frac{x^2}{2} + \mathcal{O}(x^3)$.

Homework questions

H 1 Particle in a Sphere

A particle has equal probability of being found at any point in a sphere of Radius R

- Determine the probability distribution function of the particle position in spherical coordinates $P(r, \theta, \phi)$
- Determine the probability distribution function $P(r)$ to find the particle at a radius r in the sphere
- Assume the particle is now constrained to be at a certain radius R' . What is now its probability distribution function $P(\theta, \phi|R')$?

H 2 Transformation of Probability Densities

- Explain how to transform a probability distribution function $P_X(x)$ for the random variable X , into the distribution function $P_Y(y)$ of the random variable Y , where X and Y are related by the differentiable law $y = f(x)$ in 1 dimension. Please draw a sketch explaining your intuition.
- Imagine a single molecule in an environment containing a large number of distinct binding sites. Due to the chemical/physical differences among the binding sites, their binding energy E to the molecule will be a random variable, which we assume to be normally distributed:

$$P_E(E) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(E - \mu)^2}{2\sigma^2}\right). \quad (6)$$

We take the *detachment-rate* k_{off} of the molecule from a site of energy E to be:

$$k_{\text{off}} = k e^{\beta E}, \quad (7)$$

with constants k (dimensionally $[k] = \frac{1}{\text{sec}}$) and $\beta = \frac{1}{K_B T}$ (T is absolute Temperature, K_B Boltzmann's constant). Applying the same principle as point a), compute the probability density function $P(k_{\text{off}})$ (known as **Log-normal** density function).

- Compute mean value $\langle k_{\text{off}} \rangle$. Briefly comment on the result.

H 3 Random Walker

Starting from position 0, a walker moves randomly along a line by making each second a step forward (+1) or backward (−1) with probabilities p and q respectively ($p + q = 1$). The possible positions of the walker are therefore the integers $-\infty < n < \infty$.

- Calculate the position average and the position variance after r steps. Hint: consider the Binomial distribution
- What is the probability $p_n(r)$ for the walker to be at position n after r steps? Derive $p_n(r)$:
 - for the special case $p = q = \frac{1}{2}$, by counting the number of sequences of r steps that end up in n
 - for the general case, by extracting $p_n(r)$ from the characteristic function $G_Y(k, r)$ with the stochastic variable Y indicating the position of the walker after r steps. Remember that $Y = \underbrace{X_0 + X_0 + \dots}_{r \text{ times}} \Rightarrow G_Y(k, r) = G_{X_0}^r(k)$. For Fourier inversion, use $\int_k e^{ik(a-b)} dk = \delta(a-b)$ (Dirac delta distribution).
- For case $p = q = \frac{1}{2}$, show that $p_n(r)$ can be approximated by a Gaussian distribution in the limit $r \rightarrow \infty$.
Hint: Try to find an approximation of $\ln[p_n(r)]$ in the limit of large r . Stirling formula $a! \approx \sqrt{2\pi a} \left(\frac{a}{e}\right)^a$ is also useful